Elementary Analysis

Section 34: Fundamental Theorem of Calculus

Exercises 34.1–34.12, with every subpart
Detailed worked solutions

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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34 Fundamental Theorem of Calculus

There are two complementary facts to keep straight. If f is continuous, then $H(x) = \int_a^x f(t)\,dt$ is differentiable and $H'(x) = f(x)$. If $G' = f$ and the relevant integrability hypotheses hold, then $\int_a^b f = G(b) - G(a)$. An accumulation integral can remain continuous even at a jump of its integrand, although it may then have a corner. For reversed endpoints we use $\int_b^a f = -\int_a^b f$.

34.1 Exercise 34.1: Recovering the antiderivative evaluation formula

What we must show. Deduce $\int_a^b g' = g(b) - g(a)$ when g' is continuous. Define $F(x) = \int_a^x g'(t)\,dt$ for $x \in [a, b]$. Continuity of g' guarantees that it is integrable and, by Theorem 34.3, $F'(x) = g'(x)$ for interior x. Thus $(F - g)' = 0$ on (a, b). Both F and g are continuous on the closed interval, so Corollary 29.5 (the zero-derivative consequence of the Mean Value theorem) says $F - g$ is constant there. At a, $F(a) - g(a) = -g(a)$; hence $F(b) - g(b) = -g(a)$. Rearranging gives

$$\int_a^b g'(t)\,dt = F(b) = g(b) - g(a).$$

Conclusion. The continuous-integrand version of the Fundamental Theorem implies the antiderivative evaluation formula in this case.

34.2 Exercise 34.2: Two local averages of an exponential

What we must show. Calculate the two limits using derivatives of accumulation integrals.

(a) Center the integral at zero.

Let $H(x) = \int_0^x e^{t^2}\,dt$. The integrand is continuous, so $H'(0) = e^0 = 1$ and $H(0) = 0$. By the definition of derivative,

$$\lim_{x\to 0} \frac{1}{x} \int_0^x e^{t^2}\,dt = \lim_{x\to 0} \frac{H(x) - H(0)}{x - 0} = H'(0) = 1.$$

The signed-integral convention makes this valid from both sides.

(b) Center the integral at three.

Let $K(y) = \int_3^y e^{t^2}\,dt$. Then $K(3) = 0$ and $K'(3) = e^9$. Therefore

$$\lim_{h\to 0} \frac{1}{h} \int_3^{3+h} e^{t^2}\,dt = \lim_{h\to 0} \frac{K(3+h) - K(3)}{h} = e^9.$$

Conclusion. The limits are 1 and e^9, respectively.

34.3 Exercise 34.3: A continuous accumulation function with one corner

What we must show. Integrate the piecewise function, sketch its accumulation function, and locate all derivatives. The integrand is 0 for $t < 0$, t for $0 \le t \le 1$, and 4 for $t > 1$.

(a) Integrate over the pieces actually traversed.

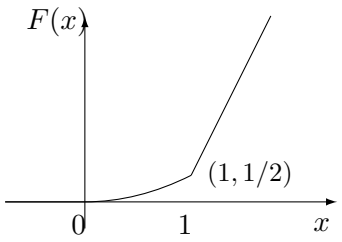
For $x \le 0$, $F(x) = \int_0^x 0\,dt = 0$. For $0 \le x \le 1$, $F(x) = \int_0^x t\,dt = x^2/2$. For $x \ge 1$, split at 1:

$$F(x) = \int_0^1 t\,dt + \int_1^x 4\,dt = \tfrac{1}{2} + 4(x-1) = 4x - \tfrac{7}{2}.$$

The isolated value of the integrand at 1 has no effect on the latter integral.

(b) Join the graph pieces at their actual heights.

The graph is horizontal at height zero to the left of 0, then follows a parabola to $(1, 1/2)$, then a straight line of slope 4. At 0 both neighboring formulas give 0; at 1 both give $1/2$. The formulas are continuous on their separate pieces, so F is continuous at every real number.



The drawing scales the horizontal coordinate by 40 and the vertical coordinate by 20, so the right line has its correct slope in data coordinates.

(c) Check the joining points separately.

Away from $0, 1$, differentiation gives $0, x, 4$ on the three open pieces. At zero, the difference quotient is 0 for negative increments and $h/2$ for positive increments, so $F'(0) = 0$. At 1 the left-hand difference quotient tends to 1 and the right-hand one tends to 4. Thus no derivative exists there.

Conclusion. F is continuous on $\mathbb{R}$ and differentiable precisely on $\mathbb{R} \setminus \{1\}$, with $F' = 0$ for $x \le 0$, $F' = x$ for $0 < x < 1$, and $F' = 4$ for $x > 1$.

34.4 Exercise 34.4: Two corners in another accumulation function

What we must show. Repeat the analysis when $f(t) = t$ for $t < 0$, $f(t) = t^2 + 1$ for $0 \le t \le 2$, and $f(t) = 0$ for $t > 2$.

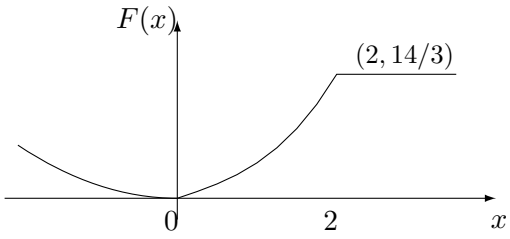
(a) Keep the orientation of the integral for negative x.

For $x < 0$, $F(x) = -\int_x^0 t\,dt = x^2/2$. For $0 \le x \le 2$, integration gives $F(x) = x^3/3 + x$. For $x > 2$, the integrand vanishes after 2, so $F(x) = F(2) = 8/3 + 2 = 14/3$. Thus

$$F(x) = \begin{cases} x^2/2, & x \le 0, \\ x^3/3 + x, & 0 \le x \le 2, \\ 14/3, & x \ge 2. \end{cases}$$

(b) Sketch and test continuity.

The left parabola decreases to $(0, 0)$; the middle cubic expression increases to $(2, 14/3)$; the remaining graph is horizontal. The two formulas at zero both give zero, and those at 2 both give $14/3$. Thus the function is continuous everywhere.



This is a schematic plot of the exact formulas; the middle curve is represented by closely spaced points.

(c) Compare slopes at the joins.

On the three open pieces the derivative is respectively x, x^2+1, and 0. At 0, the one-sided derivatives are 0 and 1. At 2, they are 5 and 0. The unequal one-sided derivatives exclude differentiability at each joining point.

Conclusion. F is continuous on all of $\mathbb{R}$, differentiable on $\mathbb{R}\setminus\{0,2\}$, and has the piecewise derivative just computed.

34.5 Exercise 34.5: Moving both endpoints

What we must show. Differentiate $F(x)=\int_{x-1}^{x+1} f(t)\,dt$ for continuous f on $\mathbb{R}$. Choose the fixed-base primitive $H(u)=\int_0^u f(t)\,dt$. The Fundamental Theorem gives $H'(u)=f(u)$ for every u. Interval additivity, with signed integrals when necessary, gives

$$F(x)=H(x+1)-H(x-1).$$

Apply the chain rule to each term. The derivatives of $x+1$ and $x-1$ are both 1, so

$$F'(x)=H'(x+1)-H'(x-1)=f(x+1)-f(x-1).$$

Conclusion. F is differentiable everywhere, with the displayed derivative.

34.6 Exercise 34.6: A trigonometric upper endpoint

What we must show. Differentiate $G(x)=\int_0^{\sin x} f(t)\,dt$ for continuous f. As above, let $H(u)=\int_0^u f(t)\,dt$, so $H'=f$ everywhere. Now $G=H\circ\sin$, and the chain rule yields

$$G'(x)=H'(\sin x)\cos x=f(\sin x)\cos x.$$

This works even when $\sin x<0$ because H uses oriented integrals, and at zeros of $\cos x$ because the chain rule includes zero inner derivatives.

Conclusion. G is differentiable on all of $\mathbb{R}$ and $G'(x)=f(\sin x)\cos x$.

34.7 Exercise 34.7: A substitution with a reversed direction

What we must show. Evaluate $\int_0^1 x\sqrt{1-x^2}\,dx$ by change of variables. Choose $u=1-x^2$, so $du=-2x\,dx$ and $x\,dx=-du/2$. As x goes from 0 to 1, u goes from 1 to 0. Keeping these transformed

endpoints gives

$$\int_0^1 x\sqrt{1-x^2}\,dx = -\frac{1}{2}\int_1^0 u^{1/2}\,du = \frac{1}{2}\int_0^1 u^{1/2}\,du = \frac{1}{2}\left[\frac{2}{3}u^{3/2}\right]_0^1 = \frac{1}{3}.$$

The substitution is justified by the change-of-variables theorem: its inner map is continuously differentiable and $\sqrt{u}$ is continuous on the transformed interval $[0,1]$. No derivative of $\sqrt{u}$ at zero is needed.

Conclusion. The value of the integral is $1/3$.

34.8 Exercise 34.8: Two choices of antiderivative in integration by parts

What we must show. Evaluate $I = \int_0^1 x\arctan x\,dx$ using two versions of v.

(a) Choose $u = \arctan x$ and $v = x^2/2$.

Then $u' = 1/(1+x^2)$ and $v' = x$. Integration by parts gives

$$I = \left[\frac{x^2}{2}\arctan x\right]_0^1 - \frac{1}{2}\int_0^1 \frac{x^2}{1+x^2}\,dx.$$

The boundary term is $\pi/8$. Since $x^2/(1+x^2) = 1 - 1/(1+x^2)$, the remaining integral is $1-\pi/4$. Therefore

$$I = \frac{\pi}{8} - \frac{1}{2}\left(1 - \frac{\pi}{4}\right) = \frac{\pi}{4} - \frac{1}{2}.$$

(b) Choose instead $v = (x^2+1)/2$.

This still has derivative x. Now the denominator cancels immediately:

$$I = \left[\frac{x^2+1}{2}\arctan x\right]_0^1 - \int_0^1 \frac{x^2+1}{2(1+x^2)}\,dx = \frac{\pi}{4} - \frac{1}{2}.$$

Both calculations are valid because all functions and derivatives used are continuous on $[0,1]$.

Conclusion. The common answer is $\pi/4 - 1/2$. A useful constant in v can simplify the remaining integral.

34.9 Exercise 34.9: Integrating the inverse sine

What we must show. Evaluate $\int_0^{1/2} \arcsin x\,dx$. The inverse-function integration identity of Example 3, or integration by parts with $u = \arcsin x$ and $v = x$, gives

$$\int_0^{1/2} \arcsin x\,dx = [x\arcsin x]_0^{1/2} - \int_0^{1/2} \frac{x}{\sqrt{1-x^2}}\,dx.$$

Here $\arcsin(1/2) = \pi/6$, so the boundary term is $\pi/12$. Since $d(\sqrt{1-x^2})/dx = -x/\sqrt{1-x^2}$,

$$\int_0^{1/2} \frac{x}{\sqrt{1-x^2}}\,dx = \left[-\sqrt{1-x^2}\right]_0^{1/2} = 1 - \frac{\sqrt{3}}{2}.$$

All derivatives in this calculation are continuous on $[0,1/2]$, which stays away from the singular endpoints ± 1.

Conclusion. $\displaystyle\int_0^{1/2} \arcsin x\,dx = \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1.$

34.10 Exercise 34.10: The areas under a function and its inverse

What we must show. Give a geometric proof that $\int_0^1 g(x)\,dx + \int_0^1 g^{-1}(u)\,du = 1$. Because g is a continuous strictly increasing bijection of $[0,1]$ onto itself, $g(0) = 0$, $g(1) = 1$, and its inverse is continuous. Consider the unit square with horizontal coordinate x and vertical coordinate u. The graph $u = g(x)$ divides it into two regions. The region below the graph has area $\int_0^1 g(x)\,dx$, by vertical slices. The complementary region is described by $g(x) \le u$; strict increase makes this equivalent to $x \le g^{-1}(u)$. Its horizontal slice at height u therefore has length $g^{-1}(u)$, so its area is $\int_0^1 g^{-1}(u)\,du$. Together the two regions fill the square, sharing only their boundary curve, so their areas sum to its area 1.

Conclusion. The two integrals are the complementary areas in the unit square.

34.11 Exercise 34.11: A continuous square with zero integral

What we must show. Show that $\int_a^b f(x)^2\,dx = 0$ forces continuous f to vanish everywhere. Suppose instead that $f(c) \neq 0$ for some $c \in [a,b]$. Continuity at c gives a relative neighborhood on which $|f(x) - f(c)| < |f(c)|/2$. The reverse triangle inequality gives $|f(x)| \ge |f(c)| - |f(x) - f(c)| > |f(c)|/2$ there. Even if c is an endpoint, this relative neighborhood contains a closed subinterval $J \subseteq [a,b]$ of positive length. Thus

$$\int_a^b f(x)^2\,dx \ge \int_J f(x)^2\,dx \ge \frac{|f(c)|^2}{4}\,|J| > 0,$$

a contradiction. The comparison is justified because f^2 is continuous and nonnegative throughout the interval.

Conclusion. No such c exists, so $f(x) = 0$ for every $x \in [a,b]$.

34.12 Exercise 34.12: Testing against every continuous function

What we must show. If $\int_a^b f(x)g(x)\,dx = 0$ for every continuous g, prove that the continuous function f is identically zero. The assumption permits *any* continuous test function g; in particular, choose $g = f$, which is continuous by hypothesis. The assumed identity becomes

$$\int_a^b f(x)^2\,dx = 0.$$

Exercise 34.11 applies directly and yields $f = 0$ everywhere. Conversely, the zero function clearly has zero integral against every continuous test function, although only the forward implication was requested.

Conclusion. f must vanish identically on $[a,b]$.